

# Results

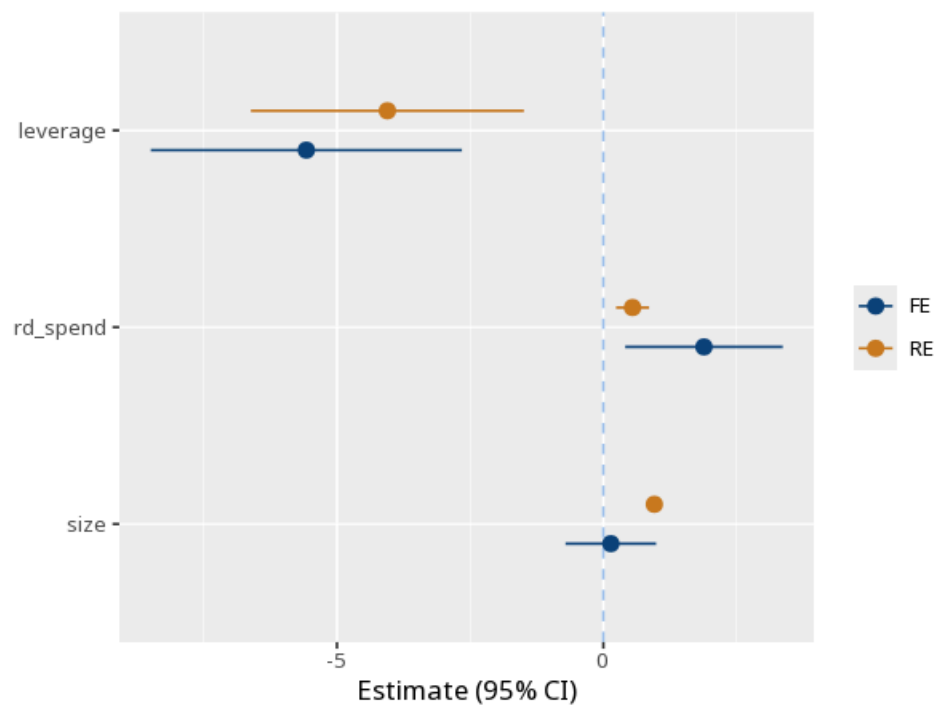
01 • HAUSMAN TEST

fixed vs. random effects?

Table 1. Hausman test

|                                     | Fixed effects                     | Random effects                    | Difference |
|-------------------------------------|-----------------------------------|-----------------------------------|------------|
| leverage                            | -5.57<br>(1.47)<br>[-8.49, -2.66] | -4.05<br>(1.29)<br>[-6.62, -1.49] | -1.52      |
| rd_spend                            | 1.89<br>(0.74)<br>[0.41, 3.37]    | 0.55<br>(0.16)<br>[0.24, 0.86]    | 1.34       |
| size                                | 0.14<br>(0.43)<br>[-0.71, 0.99]   | 0.96<br>(0.05)<br>[0.86, 1.05]    | -0.82      |
| Num.Obs.                            | 96                                |                                   |            |
| N entities                          | 12                                |                                   |            |
| R <sup>2</sup>                      | .91                               | .98                               |            |
| Hausman $\chi^2(3) = 3.07$ , p .381 |                                   |                                   |            |

Figure. FE vs. RE



How to read this test

Tests whether the random-effects assumption holds. A p below alpha means the estimates differ systematically → use fixed effects; a non-significant p means random effects is acceptable (and more efficient). Method: R plm package (Croissant & Millo, 2008). The standard errors in parentheses are clustered by entity; the bracketed line is the 95% CI. Your significance threshold ( $\alpha$ ) is 0.05.

**APA template:** A Hausman test comparing the fixed- and random-effects estimates gave  $\chi^2(3)=3.07$ ,  $p = .381$ .

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